

# Closing the Fidelity Gap: Real-Stack-Grounded Link Abstraction for 5G Digital Twins

**Abstract**—Network digital twins are essential for developing O-RAN rApps before deployment, yet open platforms force a trade-off among protocol-stack realism, site-specific propagation, network scale, and closed-loop control—none combines all four. We address the fidelity gap at its root: the link-to-system (L2S) abstraction. Instead of idealized SINR-to-throughput curves, our twin maps per-resource-element SINR to throughput through a curve measured on OpenAirInterface’s (OAI) real physical layer (nr\_dlsim with LDPC decoding), so network-scale KPIs inherit a real receiver’s implementation losses. We quantify this gap: at 10 dB SINR, the OAI curve delivers 2.16 bits/s/Hz—38% below the Shannon bound (3.46 bits/s/Hz)—and its MCS steps are non-uniform, spanning up to 2.2 dB versus the  $\sim 1$  dB of an idealized staircase. The twin builds multi-cell, multi-user 5G scenes from OpenStreetMap, ray-traces every cell-to-user channel with Sionna RT, and hosts a proportional-fair traffic-steering rApp. Two bridges connect the analytical twin to a live OAI stack: exact ray-traced channel injection and a per-UE interference-as-noise-floor mapping. Across three cities and eight random layouts each (24 runs), the rApp raises median per-user throughput by  $9.5 \pm 11.3\%$  and served cell-edge throughput by  $22.4 \pm 24.8\%$ , improves Jain fairness by 9.1%, and reduces peak cell load by 0.75 users, while leaving aggregate rate and coverage-limited outage unchanged. This gap is not cosmetic: replacing the OAI curve with an idealized one inflates throughput by 35% and changes the rApp’s steering decisions in 42% of layouts while mis-estimating its per-deployment gain by a median of 7.5 points. Finally, we close the loop on a live OAI 5G Standalone stack and measure the gap: link-level rates overpredict iperf3 goodput by a median of 53% (rApp gain gap 4.9 points on one gNB); on two real cells with scripted F1 handovers (8/8 verified) the twin’s per-UE gain (66% mean) overshoots the real gain (9.0%) by  $57 \pm 165$  points.

**Index Terms**—Digital twin, 5G NR, OpenAirInterface, link-to-system abstraction, O-RAN, rApp, traffic steering.

## I. INTRODUCTION

The digital twin is becoming the environment in which AI/ML-driven RAN control—O-RAN rApps and xApps—is trained and validated before it touches a production network [1]. Its value hinges on data fidelity: a traffic-steering rApp decides which cell serves each user from signal, interference, and load; those quantities depend jointly on 3D propagation (site-specific), on neighbouring cells (multi-cell), and on a link model that turns SINR into a real modulation-and-coding outcome (protocol-stack realism). Evaluating such a controller requires all of these together, and—because random deployments vary widely—across many layouts rather than a single scenario. A twin that reports a single headline

number from one scenario can mislead: our own single-seed pilot suggested a uniformly large rApp gain, which a 24-run study tempered into a robust-direction, variable-magnitude result.

Open tooling splits into camps that do not meet. System-level simulators scale to many cells but abstract the physical (PHY) layer with idealized SINR-to-throughput curves. OAI-based emulators execute the real 5G-NR stack but, with realistic propagation, cover a single link. Commercial twins close the rApp loop but are proprietary. No open platform combines a real protocol stack, a site-specific ray-traced channel, a multi-cell network, and a closed control loop (Table I).

A subtler gap concerns the *L2S abstraction* itself. System-level evaluation cannot afford full PHY processing per user, so it relies on a mapping from SINR to block-error-rate (BLER). Open simulators use idealized or 3GPP-calibrated curves; we obtain ours from OAI’s real receiver (nr\_dlsim with LDPC decoding and rate matching), so network-scale KPIs inherit a real implementation’s losses—particularly at MCS transition boundaries where control decisions are sensitive. At 10 dB SINR, the OAI-measured curve yields 2.16 bits/s/Hz, while the Shannon bound gives 3.46 bits/s/Hz—a 38% gap that an idealized curve would ignore (Fig. 2).

This paper contributes:

- 1) An *OAI-grounded L2S abstraction, and evidence that it matters for control*. The SINR-to-throughput curve is measured from OAI’s real PHY (38% below Shannon at 10 dB, with non-uniform MCS steps up to 2.2 dB). We then show the fidelity gap is first-order, not cosmetic: replacing this curve with an idealized one changes the traffic-steering rApp’s decisions in 42% of layouts and mis-estimates its per-deployment gain by a median of 7.5 points.
- 2) An open, CPU-only, multi-cell/multi-user digital twin: OpenStreetMap geometry  $\rightarrow$  Sionna RT ray tracing  $\rightarrow$  per-UE/cell throughput, with a swappable network source for real cell-site data.
- 3) Two twin-to-real-stack bridges: (a) exact ray-traced channel injection into a live OAI gNB–UE link, and (b) a per-UE interference-as-noise-floor mapping that lets a single-link OAI run reflect its multi-cell context.
- 4) A *measured, end-to-end fidelity gap—link and multi-cell*. On one gNB we measure deliverable iperf3 goodput against the twin’s link-level rate (median 53% inflation;

rApp per-UE gain gap  $4.9 \pm 4.3$  points). On two real cells (CU+F1 DUs) we execute the rApp's steering via scripted F1 handovers (8/8 verified) and find the twin's per-UE gain (66% mean) overshoots the real gain (9.0%) by  $57 \pm 165$  points.

- 5) A rigorous multi-scene, multi-seed evaluation (24 runs across 3 cities) of a proportional-fair traffic-steering rApp, with honest reporting of variance and outage, a load-sensitivity sweep separating coverage-limited and interference-limited regimes, and a correlation analysis showing rApp gain is limited by baseline outage rather than by raw load imbalance.

## II. RELATED WORK

**Open RAN digital-twin platforms.** Table I positions the closest open efforts on four axes. OWDT [2] pairs OAI with Sionna RT for a faithful single gNB-UE link but is single-cell and monitor-only. Tiny-Twin [3] runs a real OAI stack for multiple UEs with a RIC loop, but single-cell from replayed channel traces. Orange's RIC-TaaP/ns-O-RAN and OpenTwin [4] close rApp loops across many cells—OpenTwin adds twin-to-real KPM self-correction—but on ns-3's abstracted PHY. Recent OAI+FlexRIC work [5] closes handover loops across real cells but over the air, without a site-specific ray-traced channel. Within OAI, an emerging ray-tracing channel emulator feeds the virtual-time simulator VRTSim (multi-node on its roadmap). Commercial twins (e.g., Rimedo Labs' NDT with VIAVI) close the loop but are neither open nor reproducible.

**Link-to-system abstraction.** System-level simulation [6] relies on L2S mapping: a link-level simulator characterises SINR-to-BLER for each MCS, and an effective-SINR mapping (EESM [7], [8]) collapses a frequency-selective SINR vector into an AWGN-equivalent value. The distinction here is that our lookup is measured from OAI's own PHY rather than from idealised curves, so network-scale KPIs inherit a real receiver's implementation losses. Our multi-cell interference-as-noise-floor bridge is a site-specific generalisation of the 3GPP OFDMA channel noise generator (OCNG) [9], which loads unused resources to emulate other-cell activity; we instead derive each UE's interference from the ray-traced neighbour channels.

**O-RAN control loops.** The O-RAN architecture [10], [11] externalises RAN control: xApps on the near-RT RIC (tens of ms, via E2) and rApps on the non-RT RIC (seconds and above, via O1/A1). Traffic steering via cell-individual offsets (CIO) is a canonical rApp use case [1]. A digital twin lets such an rApp be trained and validated before deployment.

## III. DIGITAL TWIN ARCHITECTURE

The platform is a pipeline with two consumers of one physics substrate (Fig. 1): a fast analytical layer that covers every cell and UE, and a real-stack layer that grounds sampled links in OAI. Both are driven by the same ray-traced channel, and the network data source is swappable so that synthetic and real deployments run through an identical path.

TABLE I  
OPEN RAN DIGITAL-TWIN PLATFORMS ON FOUR AXES.

| Platform                   | Real stack | Site RT | Multi cell | Closed loop |
|----------------------------|------------|---------|------------|-------------|
| OWDT [2]                   | ✓          | ✓       | ×          | ×           |
| Tiny-Twin [3]              | ✓          | ×       | ×          | ✓           |
| Orange ns-O-RAN / RIC-TaaP | ×          | ✓       | ✓          | ✓           |
| OpenTwin [4]               | ×          | ×       | ✓          | ✓           |
| OAI+FlexRIC [5]            | ✓          | ×       | ✓          | ✓           |
| OAI VRTSim / RT emulator   | ✓          | ✓       | ×          | ×           |
| <b>This work</b>           | ✓          | ✓       | ✓          | ✓*          |

<sup>a</sup>replayed traces; <sup>b</sup>ns-3 abstract PHY; <sup>c</sup>over-the-air; <sup>d</sup>multi-node on roadmap; \*SA loop closed & measured on live stack (scripted F1 HO for multi-cell).

**Geometry.** A WGS84 bounding box drives an OpenStreetMap (Overpass) query; footprints are extruded into a 3D Mitsuba scene with ITU materials.

**Network.** Cells and UEs come from a swappable source: a seeded random generator ( $N$  sites  $\times$  3 sectors, uniform UE drops) for controlled experiments, or a CSV reader for real cell-site coordinates (e.g., OpenCellID). Both feed the same pipeline, so the twin can be calibrated on synthetic data and run on real site data without code changes.

**Channel.** Sionna RT [12] places one transmitter per cell (3GPP sector pattern, planar array, downtilt) and one receiver per UE and computes the channel frequency response for every cell-to-UE link over an OFDM resource grid, capturing reflections, diffractions, and scattering from the building geometry. Ray-tracing depth is 3, balancing fidelity against CPU runtime.

## IV. OAI-GROUNDED KPI ENGINE

The KPI engine is where the twin's fidelity is grounded. Let  $\mathbf{H}_{u,c}(k) \in \mathbb{C}^{N_r \times N_t}$  be the ray-traced channel of link  $c \rightarrow u$  on resource element (RE)  $k$ , and  $P_c^{\text{tx}}$  the cell transmit power. The wideband received power is  $P_{u,c}^{\text{rx}} = P_c^{\text{tx}} \|\bar{\mathbf{H}}_{u,c}\|_F^2$ , where the bar denotes averaging over antennas and REs. Each UE associates to the cell maximising received power biased by a cell-individual offset (CIO), the O-RAN control knob:

$$s(u) = \arg \max_c P_{u,c}^{\text{rx}} \cdot 10^{\text{CIO}_c/10}. \quad (1)$$

Because each cell serves one UE at a time under airtime sharing, the optimal single-stream precoder is the matched filter, so the serving per-RE gain is the largest singular value  $\sigma_{\max}$  of the serving channel. With flat per-RE transmit power  $p_c$ , a neighbour load factor  $\rho \in [0, 1]$ , and thermal noise  $N_0 = kTB \cdot \text{NF}$ , the per-RE SINR is

$$\gamma_u(k) = \frac{p_s \sigma_{\max}(\mathbf{H}_{u,s}(k))^2}{\rho \sum_{c \neq s} \frac{p_c}{N_t} \|\mathbf{H}_{u,c}(k)\|_F^2 + N_0}. \quad (2)$$

The per-RE SINRs are compressed to one effective SINR by exponential effective SINR mapping (EESM), which preserves the frequency selectivity the ray tracer produces:

$$\gamma_u^{\text{eff}} = -\beta \ln \left( \frac{1}{K} \sum_k e^{-\gamma_u(k)/\beta} \right), \quad (3)$$

with per-modulation  $\beta$  calibratable by a single scale factor. The EESM  $\beta$  values are set per modulation order (QPSK, 16QAM,

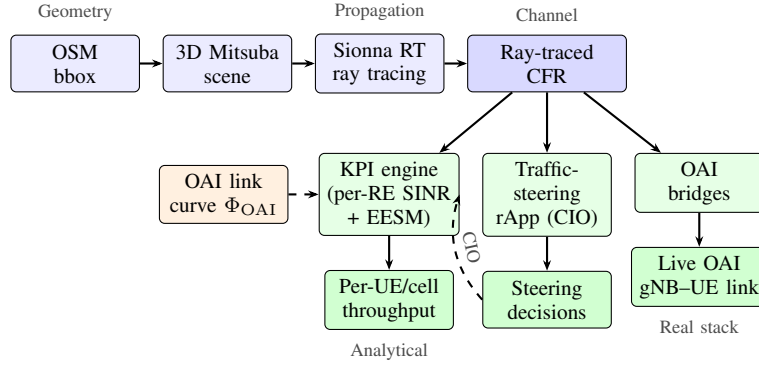


Fig. 1. Architecture. OpenStreetMap geometry and a swappable network source feed Sionna RT; the ray-traced channel drives (left) an analytical multi-cell KPI engine and the traffic-steering rApp, and (right) a live OAI link over the exact channel with a per-UE interference noise floor. The OAI-measured link curve  $\Phi_{\text{OAI}}$  grounds the analytical throughput in a real receiver.

64QAM) following established link-abstraction literature [8], and a single `eesm_beta_scale` parameter allows calibration against OAI in-the-loop ground truth.

**OAI-measured link curve.** The effective SINR is mapped to a modulation-and-coding scheme (MCS) and spectral efficiency  $\eta_u$  through a curve  $\Phi_{\text{OAI}}$  *measured offline from OAI's `nr_dlsim`* (real LDPC decoding and rate matching), i.e.  $(m_u, \eta_u) = \Phi_{\text{OAI}}(\gamma_u^{\text{eff}})$ . This is the key distinction from prior system-level twins: instead of an idealised staircase or a 3GPP-calibrated analytical expression, the SINR-to-throughput mapping carries a real receiver's implementation losses—including LDPC decoding margins, rate-matching overhead, and the non-uniform SINR spacing between MCS transitions that a real stack exhibits. The curve is measured once by sweeping  $\text{SNR} \times \text{MCS}$  through `nr_dlsim` and recording the lowest SNR at which first-transmission BLER stays within the target (10%); the resulting staircase (Fig. 2) is a monotone lookup table indexed by effective SINR.

**Quantifying the implementation loss.** Fig. 2 compares the OAI-measured curve against the Shannon bound and an idealized uniform-staircase. At 10 dB SINR, the OAI curve delivers 2.16 bits/s/Hz—38% below Shannon (3.46 bits/s/Hz). In relative terms the loss is largest at low SINR and narrows toward high SINR, but in absolute terms it grows to  $\sim 1.6$  bits/s/Hz at 64QAM, as LDPC and rate-matching overhead accumulate. Critically, the MCS transition spacing is non-uniform: the QPSK  $\rightarrow$  16QAM boundary spans 2.2 dB, while adjacent QPSK steps are  $\sim 1$  dB. An idealized staircase with uniform 1 dB steps would misplace this transition by over 1 dB, causing a system-level twin to assign the wrong MCS to boundary users—exactly the users a traffic-steering rApp aims to help.

On a static full-buffer snapshot, proportional-fair scheduling reduces to equal airtime, so the per-UE throughput is

$$R_u = \frac{B_s}{K_s} \eta_u (1 - \text{BLER}_{\text{tgt}}), \quad (4)$$

where  $B_s$  is the serving cell bandwidth and  $K_s$  its number of attached UEs.

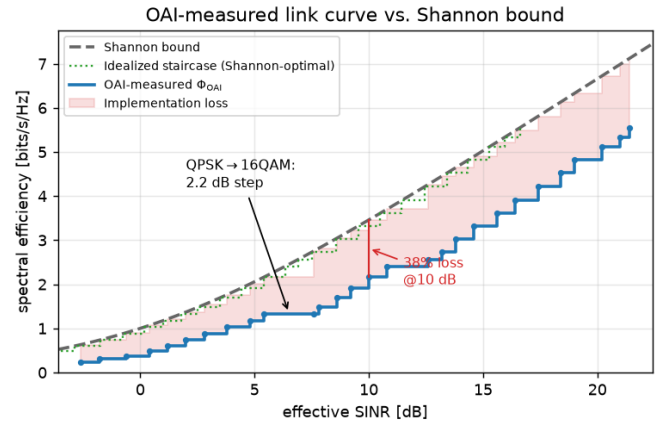


Fig. 2. OAI-measured link curve  $\Phi_{\text{OAI}}$  vs. Shannon bound and an idealized (Shannon-optimal) staircase. The shaded region is the implementation loss (38% at 10 dB). Non-uniform MCS step spacing reflects a real LDPC decoder.

#### Algorithm 1 Proportional-fair traffic steering

- 1)  $\text{CIO} \leftarrow \mathbf{0}$ ; evaluate  $R_u$  and  $U$  via Eqs. (1)–(4).
- 2) **repeat** (sweep over cells):
  - a) for each cell  $c$ : line-search  $\text{CIO}_c$  over a grid (others fixed) and keep the value maximising  $U$ .
- 3) **until** no cell's line-search improves  $U$  (local optimum).

## V. TRAFFIC-STEERING RAPP

The rApp adjusts the CIO vector to maximise the proportional-fair utility  $U(\text{CIO}) = \sum_u \log R_u$  (Alg. 1). Because  $U$  is piecewise-constant in the CIOs (it changes only when a UE's serving cell flips), a fixed small step stalls on plateaus; we therefore line-search each cell's CIO over a grid (coordinate ascent), which is monotonically non-decreasing in  $U$  by construction. Raising a cell's CIO pulls boundary UEs in (absorbing load); lowering it sheds them to neighbours.

Every KPI evaluation reuses the same engine with a `cio_db` vector, so the control loop rides on the exact same twin used for the baseline. The rApp's output is a set of per-UE handover decisions that a real stack would execute via

O1/telnet. The coordinate-ascent structure guarantees monotonic improvement in  $U$  and converges in 2–3 sweeps, suitable for the non-RT RIC timescale.

## VI. TWIN-TO-REAL-STACK BRIDGES

Two bridges connect the analytical twin to a live OAI stack, grounding per-link realism.

**Channel injection.** A bridge converts a link’s ray-traced channel frequency response to a sample-spaced impulse response (IFFT), retains the strongest causal taps, normalises them to unit energy, and maps each tap delay to a sample index. These taps override the rfsimulator’s channel through a patch to OAI’s channel module (`oai_rt_inject_channel` in `random_channel.c`). When `OAI_RT_TAPS` is set, the gNB–UE link runs over the site-specific channel. Injection is confirmed in the gNB log (`[RT] injected 4 taps ... channel_length=49`); the link then runs its PHY/MAC over the ray-traced channel.

**Interference-as-noise-floor.** A live OAI rfsimulator link is single-cell. A second bridge computes, per UE, the inter-cell interference and expresses it as a noise-floor rise  $\Delta N_u = 10 \log_{10}((I_u + N_0)/N_0)$ , where  $I_u$  is the load-scaled sum of non-serving cells’ received power. This is written as a per-client OAI noise power, so a single-link run reflects its multi-cell context—a site-specific generalisation of OCNG [9].

## VII. EXPERIMENTAL SETUP

**Scenes.** Berlin-Mitte (2,623 buildings), Paris-Opéra (121), Manhattan-Midtown (119); geometry fetched/built once per scene and reused across seeds.

**Random layouts.** Eight seeded layouts per scene—24 runs (323 s on one CPU server). Radio and topology parameters are summarised in Table II.

**Configuration sweep.** Neighbour load factor in  $\{0, 0.25, 0.5, 0.75, 1.0\}$  on a cached channel per scene.

**Metrics.** We report per-UE downlink throughput (median and mean); the *served cell-edge* rate, defined as the 5th percentile among UEs with non-zero throughput (robust to coverage holes, unlike a raw 5th percentile which collapses to zero once outage exceeds 5%); the *outage rate*, the fraction of UEs with zero throughput; Jain’s fairness index over per-UE rates; and the peak cell load (largest number of UEs on any cell). rApp gains are reported as mean  $\pm$  standard deviation over the eight seeds per scene, and pooled across all 24 runs; a gain is undefined (excluded) for a run whose baseline metric is zero.

## VIII. RESULTS

### A. Traffic-steering rApp

Across 24 runs, relative to strongest-cell association (Table III, Fig. 3): the rApp helps the users a strongest-cell policy under-serves (median, cell-edge, fairness) and sheds peak load. In absolute terms the overall median per-UE rate rises from 8.5 to 9.2 Mbps and the served cell-edge rate from 3.5 to 4.2 Mbps, while the mean is essentially unchanged (11.4 Mbps). Three

TABLE II  
RADIO AND TOPOLOGY PARAMETERS.

| Parameter                      | Value                 |
|--------------------------------|-----------------------|
| Carrier / bandwidth            | 3.5 GHz / 20 MHz      |
| Subcarrier spacing             | 30 kHz (numerology 1) |
| Cell transmit power            | 46 dBm                |
| BS array / UE antennas         | $4 \times 1 / 1$      |
| Receiver noise figure          | 7 dB                  |
| Ray-tracing interaction depth  | 3                     |
| Cells / UEs per layout         | 9 / 30                |
| Default inter-cell load $\rho$ | 1.0                   |

TABLE III  
RAPP GAINS VS. STRONGEST-CELL (MEAN  $\pm$  STD OVER SEEDS).

| Metric             | Overall           | Berlin | Paris | Manh. |
|--------------------|-------------------|--------|-------|-------|
| Median tput gain   | $9.5 \pm 11.3\%$  | 10.5%  | 4.5%  | 13.3% |
| Served edge gain   | $22.4 \pm 24.8\%$ | 41.0%  | 6.0%  | 20.2% |
| Jain fairness gain | $9.1 \pm 11.0\%$  | 11.2%  | 4.4%  | 11.5% |
| Peak load (UEs)    | −0.75             | −0.88  | −0.50 | −0.88 |
| Mean tput gain     | $0.5 \pm 5.8\%$   | −0.7%  | −0.3% | 2.5%  |
| Outage change      | 0.0 pts           | 0.0    | 0.0   | 0.0   |

honest observations: aggregate rate barely moves (+0.5%; the proportional-fair objective trades sum for fairness), outage is unchanged (steering relocates covered users, it does not create coverage), and variance is high (layout-dependent)—motivating multi-seed evaluation.

The per-scene breakdown is instructive (Table III). Berlin, the densest scene, shows the largest served cell-edge gain (+41%); dense blockage produces stronger load imbalance for the rApp to exploit. Paris shows the smallest gain (+6%) and the highest outage (23.8%): when many users lack coverage entirely, steering has little covered load to rebalance. Manhattan sits in between. Across all three cities the direction is robust—median, cell-edge, and fairness improve while peak load falls—while the magnitude is strongly deployment-dependent, exactly what a single-scenario study would misrepresent.

### B. The fidelity gap changes rApp evaluation

The 38% link-curve gap of Sec. IV is not cosmetic—it changes what the twin tells an rApp developer. We re-run the *identical* rApp on a twin whose *only* change is the link curve:  $\Phi_{\text{OAI}}$  is replaced by an idealized curve that assigns the same MCS set at the Shannon-optimal SINR (no implementation loss), with the ray-traced channel, association, and interference held fixed. As expected, the idealized twin inflates baseline median throughput by 35%. More consequentially, it *disagrees with the real-PHY twin on the control decision*: across the 24 layouts its predicted rApp median gain deviates from the OAI-grounded value by a median of 7.5 (mean 8.2, max 29) points, the two agree only moderately ( $r=0.59$ ; Fig. 4), and in 10/24 layouts (42%) the rApp selects a *different* set of steering actions (mean steered-user Jaccard 0.75). Yet the *ensemble-average* gain is nearly identical (10.9% vs 10.4%): an idealized twin looks calibrated in aggregate while being unreliable per

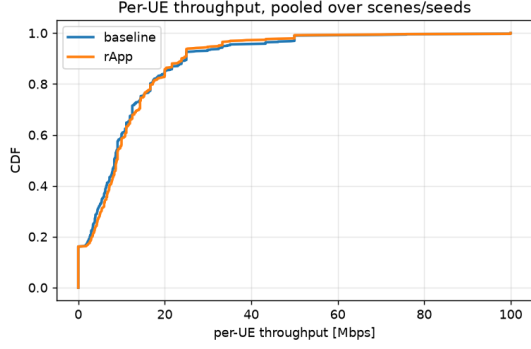


Fig. 3. Per-UE throughput CDF, baseline vs. rApp (pooled over all scenes and seeds). The rApp lifts the low-to-median part of the distribution.

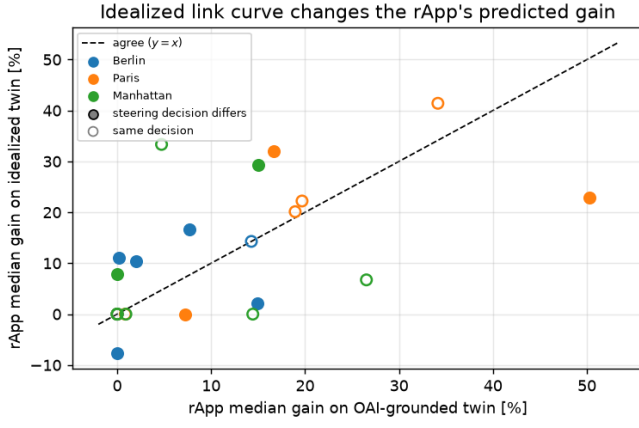


Fig. 4. Predicted rApp median gain on an idealized-curve twin vs. the OAI-grounded twin (24 layouts). Points off  $y=x$  show the idealized twin mis-estimating the gain; filled markers are the 10/24 layouts where it also selects different steering actions.

deployment—exactly the regime in which an operator tunes and ranks a policy for a specific site. This is the concrete cost of the fidelity gap, and why a control-oriented twin must be grounded in a real receiver.

### C. The real fidelity gap

The analysis above compares two *models*; we now close the loop on the real stack and measure the gap directly. We stand up a full OAI 5G Standalone system on the same CPU host—the OAI 5G core (AMF/SMF/UPF and supporting NFs, all healthy), a gNB, and a real UE that registers, establishes a PDU session (DL IP over the UPF), and carries user-plane traffic—and confirm end-to-end connectivity by iperf3 to the data network. On a single gNB we ground the *link* dimension: we pin the downlink operating point by capping the scheduler MCS over a clean channel, sweep the cap across MCS 2–28, and measure end-to-end iperf3 goodput at each fixed MCS, which indexes the same OAI curve the twin uses (a shared SINR axis).

The result is Fig. 5 (left). The twin’s link-level rate  $B\eta_{\text{OAI}}(1 - \text{BLER})$  overpredicts real end-to-end goodput

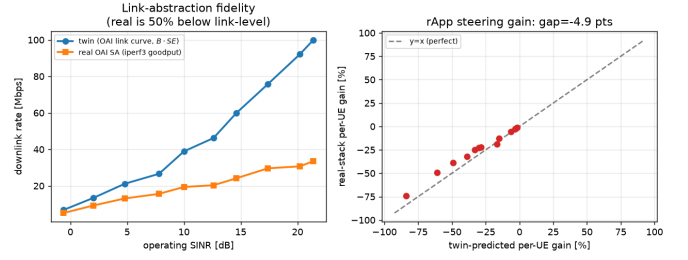


Fig. 5. Measured real fidelity gap (single gNB). Left: twin link-level rate vs. real OAI Standalone end-to-end goodput at matched operating SINR—real is a median 53% lower. Right: per-UE rApp steering gain, twin-predicted vs. real-stack (8 layouts); gap 4.9 pts.

by a median of 53% (mean 50%)—e.g. at MCS 28, 100 vs. 33.5 Mbps. This gap is *larger* than the link-model gap of Sec. IV and distinct in origin:  $\Phi_{\text{OAI}}$  is measured on an isolated, full-buffer `nr_dlsim` link, whereas the real stack pays for the TDD frame structure (7D:2U), HARQ, RLC/PDCP/SDAP/GTP headers, scheduler dynamics, and TCP. Propagated through the rApp over eight layouts (240 per-UE evaluations, 13 steered UEs), the twin-predicted per-UE steering gain (−28.5% mean) and the real gain (−23.7%) differ by  $4.9 \pm 4.3$  points (Fig. 5, right).

### D. Multi-cell closed loop on the real stack

Two real cells (CU plus F1 DUs, PCI 0/1) match a two-sector twin (20 UEs, full neighbour load). On eight Berlin layouts where the rApp moves a UE, we measure iperf3 goodput on the baseline and steered cells and execute the move via scripted CU telnet (`ci trigger_fl_ho`; all 8/8 succeed with PDU preserved). Fig. 6: twin mean per-UE gain 66% vs. real 9.0% (gap  $57 \pm 165$  pts). The clean rfsim channel does not reproduce the ray-traced SINR split driving association, so the twin often predicts large gains where the real move hurts—a larger, more variable gap than the single-link calibration.

### E. Coverage, outage, and load sensitivity

Mean UE outage is 16.2% at full load (Berlin 8.8%, Manhattan 16.3%, Paris 23.8%); rApp median gain declines with baseline outage ( $r=-0.42$ ) more than with load imbalance ( $R^2=0.04$ ). Lowering neighbour load from 1.0 to 0 raises mean throughput  $1.7\text{--}1.8\times$ , separating a coverage-limited tail steering cannot fix from an interference-limited body where load-balancing acts.

## IX. DISCUSSION AND LIMITATIONS

We state these plainly. (1) The OAI Standalone loop is closed on both one gNB (link calibration, Sec. VIII-C) and two cells with scripted F1 handovers (Sec. VIII-D); autonomous E2-RC and per-UE ray-traced channels on the multi-cell rfsim stack remain future work. (2) Inter-cell interference is modelled as a noise-floor rise, not physically combined waveforms (which need FPGA/GPU emulators). (3) The evaluation is a static full-buffer snapshot, single carrier, downlink;

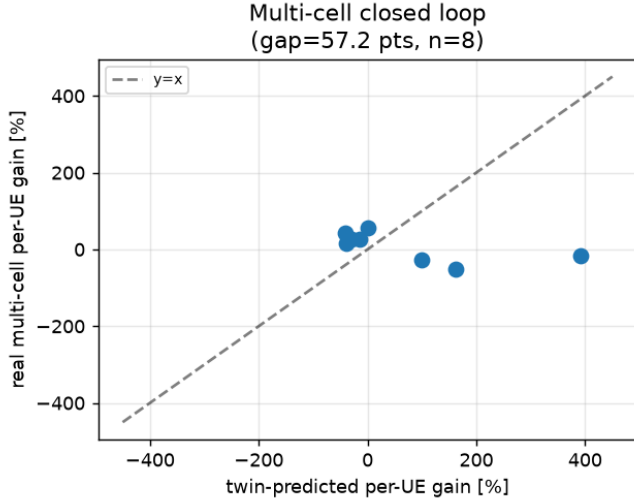


Fig. 6. Multi-cell closed loop (8 layouts, 8/8 handovers verified). Twin-predicted vs. real per-UE rApp steering gain; points off  $y=x$  show the twin mis-estimating delivered gain (mean gap 57 pts).

mobility, FFR, and uplink are natural extensions. (4) rApp benefits are high-variance and layout-dependent (we report full distributions), and steering does not address coverage. (5) Validation is stack-versus-model; field calibration and real cell-site databases are the next steps.

## X. CONCLUSION

We presented an OAI-grounded, site-specific multi-cell twin with traffic-steering rApp, two real-stack bridges, and rigorous multi-scene evaluation. Replacing the OAI curve with an idealized one changes steering in 42% of layouts (median gain error 7.5 points). Closing the loop on the real OAI stack—single gNB and two-cell CU/F1—shows deliverable goodput is a median 53% below link-level prediction; on two cells the twin’s per-UE gain (66%) overshoots the real gain (9.0%) by 57 points. An OAI-grounded twin must still be closed against the real stack before its control conclusions can be trusted. Code and `experiments/*_fidelity_gap.py` reproduce all results (Apache-2.0).

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